A Sum-of-Squares Decomposition for Quartet Covariance: Bivariate Independence and Quartet Distance

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Abstract

We establish a nonnegative sum-of-squares representation for a quartet covariance associated with pairs of finite labelled trees. The representation has two distinct applications. First, combined with an exact quartet-counting identity, it gives $d_Q(\mathcal{T}_1, \mathcal{T}_2) \leq \frac{2}{3}\binom{n}{4} + \binom{n}{3} + \frac{1}{6}\binom{n}{2}$ for binary phylogenetic trees on the same n-taxon label set, providing an independent sum-of-squares derivation of the Bandelt–Dress asymptotic 2/3 bound. Second, after specialisation to labelled path trees, rational approximation, and nested quantisation, the same representation yields $\tau^*(X, Y) \geq 2\mathcal{B}(X, Y)$ for every real-valued bivariate law, where τ^* is the unnormalised Bergsma–Dassios sign covariance and $\mathcal{B}$ is the unscaled Blum–Kiefer–Rosenblatt functional. Consequently, $\tau^*(X, Y) = 0$ if and only if X and Y are independent, without continuity or absolute-continuity assumptions and hence for atomic, mixed, and singular laws. On finite uniformly weighted label sets, the framework also identifies an edge-weighted quartet quantity with 3/2 times empirical distance covariance squared for the induced tree pseudometrics. The two applications use different consequences of the same finite-tree identity: full nonnegativity in the quartet-distance argument and a coefficient estimate in the bivariate-law argument.

Keywords: quartet covariance, sum-of-squares decomposition, labelled trees, quartet distance, Bergsma–Dassios sign covariance, singular bivariate laws

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1 Introduction

Quartet patterns provide a common finite language for comparing labelled trees and for studying dependence in bivariate laws. Four sampled labels induce partial quartets in each of two trees, and agreement or disagreement between the induced patterns defines a signed quartet covariance. Our starting point is a structural identity for this covariance: for every pair of finite X-trees, it decomposes into four families of nonnegative squares indexed by local edge and length-2 path configurations. This algebraic–combinatorial decomposition holds for finite labelled trees of arbitrary topology; path trees and binary phylogenetic trees enter only in the applications.

1.1 Two principal applications

The first concerns quartet distance. The full nonnegativity of the sum-of-squares representation, combined with an exact counting identity, gives $d_Q(\mathcal{T}_1, \mathcal{T}_2) \leq \frac{2}{3}\binom{n}{4} + \binom{n}{3} + \frac{1}{6}\binom{n}{2}$ for two binary phylogenetic trees on n taxa. This provides an independent sum-of-squares derivation of the asymptotic $2/3$ upper bound conjectured by Bandelt and Dress [1].

The second application concerns Bergsma–Dassios sign covariance τ^*. A coefficient estimate extracted from the first square family, specialised to labelled path trees, yields $\tau^*(X, Y) \geq 2\mathcal{B}(X, Y)$, where $\mathcal{B}$ is the unscaled Blum–Kiefer–Rosenblatt functional [2]. Rational approximation and nested quantisation extend the finite path-tree inequality to every real-valued bivariate law. Consequently, Corollary 3 proves that $\tau^*(X, Y) = 0$ if and only if X and Y are independent, including for atomic, mixed, and singular laws and without density, continuity, or moment assumptions. The two applications therefore use different consequences of one finite-tree identity: full nonnegativity in the quartet-distance argument and a coefficient lower bound in the bivariate-law argument. Neither conclusion is used to prove the other.

The framework has a further probabilistic interpretation. On a finite uniformly weighted label set, Theorem 3 identifies an edge-weighted quartet quantity with $3/2$ times empirical distance covariance squared for the tree pseudometrics induced by the two labelled trees. Thus quartet covariance links a local decomposition on trees, an extremal tree-comparison problem, and dependence functionals of joint laws.

1.2 Passage from finite trees to arbitrary laws

For the Bergsma–Dassios application, a finite ordered bivariate distribution whose joint point masses $P(X = x_i, Y = y_j)$ are rational is encoded by two labelled path trees, with normalised label multiplicities equal to those masses. The coefficient estimate then gives the finite-support inequality. Rational approximation handles arbitrary finite tables. Finally, right-closed nested quantisers, continuity from above, and dominated convergence transfer the inequality to arbitrary real-valued bivariate Borel laws. The bounded weak-order kernels retain ties, so atomic, mixed, and singular components require no separate regularity assumption. This separates the finite algebraic–combinatorial identity from the measure-theoretic passage to general bivariate laws.

1.3 Relation to dependence measures

Bergsma and Dassios [3] introduced τ^* as a rank-based population measure of dependence. They proved its nonnegativity and zero characterisation when the joint law is discrete, jointly absolutely continuous, or a mixture of those types, and conjectured the conclusion for arbitrary bivariate laws. Hoeffding's rank test uses patterns of five observations [4], whereas Yanagimoto [5], under the continuity assumptions of his paper, presented a four-observation measure of association. Drton, Han, and Shi [6] identified that measure, for absolutely continuous pairs, as proportional to τ^* and proved the zero characterisation for random vectors with continuous margins, including joint laws that need not be absolutely continuous. In the no-ties sample setting they also related the associated U-statistics for Hoeffding's D, the Blum–Kiefer–Rosenblatt statistic R, and τ^*. Large-sample theory and broader pattern-based frameworks were subsequently developed in [7–9]. These sample-level results are complementary to the population inequality proved here; no new sample-level limit theorem is claimed.

1.4 Relation to quartet distance

Bandelt and Dress [1] conjectured the asymptotic 2/3 upper bound for quartet distance. Alon, Naves, and Sudakov [10] established the general upper bound $0.69\binom{n}{4} + o(n^4)$. Pachter's 2026 preprint (arXiv:2608.03542), posted before the present manuscript, proves the same asymptotic 2/3 conclusion by a different route based on common-root planarisation and a five-leaf identity. The contribution here is an independent proof based on the quartet-covariance sum-of-squares representation and the explicit finite-n inequality above. An earlier version of the tree-decomposition approach and its application to the quartet-distance problem appeared in Huang's 2017 doctoral dissertation [11]. The present paper gives a revised, self-contained sum-of-squares formulation and develops its connection with bivariate dependence functionals.

Section 2 introduces labelled trees, partial quartets, and quartet covariance. Section 3 proves the inclusion–exclusion and sum-of-squares identities, derives the quartet-distance application, and establishes the coefficient estimate used for path trees. Section 4 develops the normalised tree quantities and the distance-covariance identity, and proves the arbitrary-law result for τ^* in Subsection 4.3.

2 Tree and quartet covariance framework

We follow Semple and Steel [12] for standard terminology and notation concerning phylogenetic X-trees, splits, and displayed quartets.

For a vertex v of a graph G, let $N(v)$ denote the set of neighbours of v and let $N_+(v) = N(v) \cup \{v\}$. For $w \in N(v)$, define $N_v(w) := N(w) \setminus \{v\}$ and $N_{v+}(w) := N_v(w) \cup \{w\}$. Thus, for every ordered adjacent pair (t, u) in G, $N_{t+}(u) = N_t(u) \cup \{u\}$. For such an ordered adjacent pair, define the following two sets of ordered vertex pairs:

$$I_{t,u} := \{(u', u'') \in N_{t+}(u) \times N_+(u) : u' = u \text{ or } u'' \notin \{t, u'\}\}$$

and

$$I'_{t,u} := \{(u', u'') \in N_{t+}(u) \times N_+(u) : u' = u \text{ or } u'' \neq u'\}.$$

For a finite nonempty set X, an X-tree $\mathcal{T} = (T, \phi)$ is a finite (graph-theoretic) tree $T = (V, E)$ together with a labelling map $\phi : X \to V$, where $\phi^{-1}(v) \neq \varnothing$ for every vertex v of degree at most 2. An X-tree is called *trivial* if its underlying tree has no edge, and *non-trivial* otherwise. A trivial X-tree displays no partial quartet. An X-tree is called *phylogenetic* if ϕ is a bijection from X onto the set of graph-theoretic leaves of T (vertices of degree 1); thus a phylogenetic X-tree cannot have vertices of degree 2. A phylogenetic tree is called *binary* if all nonleaf vertices have degree 3.

A *partial split* of X is an unordered pair of non-overlapping non-empty subsets of X. A partial split $\{X_1, X_2\}$, with $X_1 = \{x_1^1, \ldots, x_1^i\}$ and $X_2 = \{x_2^1, \ldots, x_2^j\}$, is denoted by $X_1 \mid X_2$ or $x_1^1 \cdots x_1^i \mid x_2^1 \cdots x_2^j$. A partial split $X_1 \mid X_2$ is called a *full split*, or simply a *split*, if $X_1 \cup X_2 = X$; it is called a *partial quartet* if $|X_1|, |X_2| \leq 2$, and a *(full) quartet* if $|X_1| = |X_2| = 2$. Partial quartets $x_1 \mid x_2$ and $x_1 \mid x_2x_3$ will sometimes be denoted by $x_1x_1 \mid x_2x_2$ and $x_1x_1 \mid x_2x_3$, respectively.

For two partial splits $X_1 \mid X_2$ and $X_3 \mid X_4$ of X, we say that $X_1 \mid X_2$ *displays* $X_3 \mid X_4$ if either $X_3 \subseteq X_1$ and $X_4 \subseteq X_2$, or $X_3 \subseteq X_2$ and $X_4 \subseteq X_1$. For an X-tree $\mathcal{T} = (T, \phi)$ and an edge uv of T, let $V^u(v)$ be the set of vertices in the same component of $T - uv$ as v, and let

$$S_{uv} := \bigcup_{v' \in V^u(v)} \phi^{-1}(v') \,\Big|\, \bigcup_{u' \in V^v(u)} \phi^{-1}(u')$$

be the corresponding split of X. For $w \in N_u(v)$, let

$$S_{u,w} := \bigcup_{u' \in V^v(u)} \phi^{-1}(u') \,\Big|\, \bigcup_{w' \in V^v(w)} \phi^{-1}(w')$$

be the corresponding partial split of X. Here $u-v-w$ is the unique length-2 path: the middle vertex v, its labels, and all branches at v other than the two endpoint branches are omitted from the two displayed sides.

When H is an edge uv, write $S_H := S_{uv}$; when H is the length-2 path $u - v - w$, write $S_H := S_{u,w}$. Thus S_e for an edge e always means its edge split, whereas S_H for a length-2 path means its endpoint partial split. In later formulas whose surrounding indices specify that u, w are the distance-2 endpoints of $u - v - w$, the shorthand S_{uw} also denotes this endpoint partial split $S_{u,w}$; the distance-2 condition distinguishes this usage from the edge-split notation.

Every component of a finite tree obtained by deleting an edge contains a leaf of the original tree. Such a leaf has a nonempty label fibre by the definition of an X-tree. Consequently both sides of every edge split are nonempty. Likewise, the union of the label fibres in the component reached through any neighbouring branch is nonempty; the label fibre at the central vertex itself may be empty.

A partial split $X_1 \mid X_2$ is *displayed* by $\mathcal{T}$ if there is an edge uv of T such that S_{uv} displays $X_1 \mid X_2$. For $i \in \{2, 3, 4\}$, let $Q_i(\mathcal{T})$ denote the set of all displayed partial quartets containing precisely i taxa. Associate with a partial quartet $q = A \mid B$ a four-element multiset $M(q)$ by listing each member of a two-element side once and, for every singleton side, listing its unique taxon twice. A quadruple (x_1, x_2, x_3, x_4)

induces q if its multiset of entries is $M(q)$. Here the support of a quadruple is the set of its distinct entries. Thus a support-2 quadruple induces a partial quartet precisely for multiplicities $2+2$, and a support-3 quadruple does so precisely for multiplicities $2+1+1$; in either case the induced object is unique. A support-4 quadruple induces each of the three possible full quartet partitions on that support, while multiplicities $3+1$ or 4 induce none.

Let $\mathcal{T}_1 = (T_1, \phi_1)$ and $\mathcal{T}_2 = (T_2, \phi_2)$ be two X-trees with $T_1 = (V_1, E_1)$ and $T_2 = (V_2, E_2)$. For $i \in \{1, 2\}$, let D_i be the set of ordered pairs of adjacent vertices in T_i, let E_i^2 be the set of unordered pairs of vertices at distance 2, and let

$$D_i^2 := \{(u, v, w) \in V_i^3 : uv, vw \in E_i \text{ and } u \neq w\}$$

be the set of ordered paths of length 2 in T_i. Fix an arbitrary total order $<_i$ on V_i and put

$$P_i := \{(u, v, w) \in D_i^2 : u <_i w\}.$$

Thus P_i contains exactly one representative of each reversal orbit $\{(u, v, w), (w, v, u)\}$. Reversing the endpoints leaves the endpoint partial split unchanged. A summation condition $uvw \in E_i^2$ is shorthand for $\{u, w\} \in E_i^2$, with v the unique middle vertex; it displays the middle vertex without changing E_i^2 from a set of unordered endpoint pairs.

Let $(t, u) \in D_1$ and $(v, w) \in D_2$. Throughout this paragraph, neighbour and component notation involving t, u is taken in T_1, while that involving v, w is taken in T_2. Define $R_{t,u}(u) = \phi_1^{-1}(u)$. For $u' \in N(u)$, let $R_{t,u}(u')$ denote the union of the label sets $\phi_1^{-1}(u'')$ over all $u'' \in V_1^u(u')$. Correspondingly, define $C^{v,w}(w) = \phi_2^{-1}(w)$, and for $w' \in N(w)$ let $C^{v,w}(w')$ be the union of $\phi_2^{-1}(w'')$ over all $w'' \in V_2^w(w')$. For $u' \in N_+(u)$ and $w' \in N_+(w)$, put

$$A_{t,u}^{v,w}(u', w') := R_{t,u}(u') \cap C^{v,w}(w').$$

We call the sets $R_{t,u}(u')$ the row blocks, the sets $C^{v,w}(w')$ the column blocks, and their intersections $A_{t,u}^{v,w}(u', w')$ the cells of the resulting row–column table. For the block sets $R_{t,u}(u')$, $C^{v,w}(w')$, and $A_{t,u}^{v,w}(u', w')$ just defined, and for their later indexed variants, the corresponding lowercase letter denotes normalized cardinality, namely cardinality divided by $|X|$; for example, $r_{t,u}(u') = \frac{|R_{t,u}(u')|}{|X|}$. For every $u' \in N(u)$ and $w' \in N(w)$, the sets $R_{t,u}(u')$ and $C^{v,w}(w')$ are the nonempty branch-component label sets described above. Consequently, $r_{t,u}(u') > 0$ for $u' \in N(u)$, and $c^{v,w}(w') > 0$ for $w' \in N(w)$. These are precisely the neighbouring-branch masses used as row sums or denominators in the coefficient calculations below. No positivity is asserted for the possibly empty central fibres $R_{t,u}(u)$ and $C^{v,w}(w)$.

The coefficient family denoted by lowercase q below is unrelated to the temporary letter q used above for a generic partial quartet. For $(t, u) \in D_1$, $(v, w) \in D_2$, $u' \in N_{t+}(u)$, $u'' \in N_+(u)$, $w' \in N_{v+}(w)$, and $w'' \in N_+(w)$, define

$$q_{t,u}^{v,w}(u', u'', w', w'') := \frac{1}{2} c^{v,w}(w'') \left(a_{t,u}^{v,w}(u', w') \frac{a_{t,u}^{v,w}(u'', v)}{c^{v,w}(v)} + a_{t,u}^{v,w}(u'', w') \frac{a_{t,u}^{v,w}(u', v)}{c^{v,w}(v)} \right).$$

If $u' \in N_t(u)$, then

$$q^{v,w}_{t,u}(u', t, w', w'') = q^{v,w}_{u',u}(t, u', w', w''),$$

and, retaining the two endpoints t, u' of the path $t - u - u'$, we define

$$b^{v,w}_{tu'}(w', w'') := q^{v,w}_{t,u}(u', t, w', w'').$$

Correspondingly, $w' \in N_v(w)$ implies

$$q^{v,w}_{t,u}(u', u'', w', v) = q^{w',w}_{t,u}(u', u'', v, w'),$$

and we define

$$s^{vw'}_{t,u}(u', u'') := q^{v,w}_{t,u}(u', u'', w', v).$$

If $u' \in N_t(u)$ and $w' \in N_v(w)$, define $p^{vw'}_{tu'} := q^{v,w}_{t,u}(u', t, w', v)$. Thus each displayed b-, s-, or p-coefficient is obtained from the displayed q-coefficient by imposing, respectively, $u'' = t$, $w'' = v$, or both, at the stated indices and then using the displayed orientation identity. These are indexed coefficient identities, not identities between complete coefficient families. The lowercase coefficient p is unrelated to the path representative sets P_i. Then

$$\begin{aligned} p^{vw'}_{tu'} &= q^{v,w}_{u',u}(t, u', w', v) = q^{w',w}_{t,u}(u', t, v, w') = q^{w',w}_{u',u}(t, u', v, w') \\ &= b^{v,w}_{tu'}(w', v) = b^{w',w}_{tu'}(v, w') = s^{vw'}_{t,u}(u', t) = s^{vw'}_{u',u}(t, u'). \end{aligned}$$

The preceding identities show that b and s are unchanged by reversal of the associated path endpoints, and that p is unchanged by either reversal. The corresponding squared linear forms in Theorem 2 are also unchanged. Hence the sums over P_i do not depend on the auxiliary total order $<_i$.

Throughout this finite-label framework, drawing taxa with replacement means independent uniform draws from X; consequently every ordered quadruple has probability $|X|^{-4}$. In the statistical application, a rational probability table is represented by replicating labels so that these uniform label frequencies equal the cell probabilities.

We define the *quartet covariance* $Q(\mathcal{T}_1, \mathcal{T}_2)$ to be the probability of drawing four taxa with replacement that induce a partial quartet displayed by both trees, minus half the probability of drawing four taxa that induce one quartet displayed by $\mathcal{T}_1$ and a different quartet displayed by $\mathcal{T}_2$.

For two partial X-splits $S_1 = A \mid B$ and $S_2 = C \mid D$, define $Q(S_1, S_2)$ analogously. Then

$$Q(S_1, S_2) = \frac{6}{|X|^4} \left(\det \begin{pmatrix} |A \cap C| & |A \cap D| \\ |B \cap C| & |B \cap D| \end{pmatrix} \right)^2.$$

We call S_1, S_2 *independent* if $Q(S_1, S_2) = 0$, and call $\mathcal{T}_1, \mathcal{T}_2$ independent if S_{tu}, S_{vw} are independent for every $tu \in E_1$ and $vw \in E_2$.

For a four-element subset $Y \subseteq X$, an X-tree displays at most one full quartet with support Y. Define

$$d_Q(\mathcal{T}_1, \mathcal{T}_2) := \#\{Y \subseteq X : |Y| = 4,\ \mathcal{T}_1 \text{ and } \mathcal{T}_2 \text{ display distinct full quartets on } Y\}.$$

For binary phylogenetic trees every four-set supports exactly one displayed quartet, so this definition agrees with the usual quartet distance [13].

Finally, let

$$X_{t,u}^{v,w} := a_{t,u}^{v,w}(t,v)a_{u,t}^{w,v}(u,w) - a_{t,u}^{w,v}(t,w)a_{u,t}^{v,w}(u,v).$$

Then

$$X_{t,u}^{v,w} = -X_{t,u}^{w,v} = -X_{u,t}^{v,w} = X_{u,t}^{w,v}, \qquad Q(S_{tu}, S_{vw}) = 6\big(X_{t,u}^{v,w}\big)^2.$$

By a *pure-square term* we mean a term proportional to the square of a single $X_{t,u}^{v,w}$ after one of the squared linear forms in Theorem 2 is expanded. An individual pure-square occurrence associated with one endpoint orientation need not be invariant. Whenever a representative in P_i is used, the formulas below include the pure-square occurrences associated with both endpoint orientations.

3 Sum-of-squares decomposition, quartet distance, and coefficient bound

Using the inclusion–exclusion principle, we can compute the quartet covariance between two X-trees in terms of the splits and partial splits corresponding to edges and paths of length 2.

Theorem 2 gives the representation used below. For the statistical application, we need Lemma 6, which is obtained by retaining the first nonnegative sum and gives the local quantitative bound later applied to path trees in the proof of Corollary 3. The nonnegativity of the full identity also gives the quartet-distance bound in Corollary 2.

Lemma 1 *Let $q = A \mid B$ be a partial quartet displayed by an X-tree $\mathcal{T}$. The edges e for which S_e displays q are the edges of one nonempty path. If that path has k edges, then the length-2 paths whose endpoint partial split displays q are exactly its $k-1$ consecutive edge pairs.*

Proof Let H_A and H_B be the minimal connected subtrees spanning the labelled vertices of the taxa in A and B, respectively. Since some edge split displays $A \mid B$, these two subtrees are disjoint. An edge separates all of A from all of B exactly when it lies on the unique path joining H_A to H_B. Two such separating edges are adjacent exactly when their outer endpoint branches define one of the partial splits $S_{u,w}$ above. The last assertion is therefore the elementary count of consecutive edge pairs in a path. □

Theorem 1 *For every two X-trees $\mathcal{T}_1$ and $\mathcal{T}_2$, we have*

$$\begin{aligned} Q(\mathcal{T}_1, \mathcal{T}_2) = &\sum_{\substack{tu \in E_1 \\ vw \in E_2}} Q(S_{tu}, S_{vw}) - \sum_{\substack{tu \in E_1 \\ v_1 w v_2 \in E_2^2}} Q(S_{tu}, S_{v_1 v_2}) \\ &- \sum_{\substack{t_1 u t_2 \in E_1^2 \\ vw \in E_2}} Q(S_{t_1 t_2}, S_{vw}) + \sum_{\substack{t_1 u t_2 \in E_1^2 \\ v_1 w v_2 \in E_2^2}} Q(S_{t_1 t_2}, S_{v_1 v_2}). \end{aligned}$$

Proof By Lemma 1, for every displayed partial quartet q_i the number of displaying length-2 paths is one less than the number of displaying edges.

Let (x_1, x_2, x_3, x_4) induce a partial quartet q_i displayed by $\mathcal{T}_i$, for $i \in \{1, 2\}$. Let H_i be an edge or a path of length 2 of T_i such that S_{H_i} displays q_i. The contribution of the quadruple to both $Q(\mathcal{T}_1, \mathcal{T}_2)$ and $Q(S_{H_1}, S_{H_2})$ is $|X|^{-4}$ if $q_1 = q_2$, and $-(2|X|^4)^{-1}$ otherwise. Denote this contribution by $\gamma(x_1, x_2, x_3, x_4)$, and let k_i be the number of edges $u_i v_i$ of T_i for which $S_{u_i v_i}$ displays q_i. The contribution to the right-hand side is

$$\gamma(x_1, x_2, x_3, x_4)\big(k_1 k_2 - k_1(k_2 - 1) - (k_1 - 1)k_2 + (k_1 - 1)(k_2 - 1)\big) = \gamma(x_1, x_2, x_3, x_4).$$

Summing over all ordered quadruples proves the formula. □

We will show that the quartet covariance is a sum of squares.

Theorem 2 *For every two X-trees $\mathcal{T}_1$ and $\mathcal{T}_2$,*

$$\begin{aligned}
\frac{1}{6}Q(\mathcal{T}_1, \mathcal{T}_2) = &\sum_{\substack{(t,u)\in D_1\\(v,w)\in D_2}} \sum_{\substack{(u',u'')\in I_{t,u}\\(w',w'')\in I_{v,w}}} q_{t,u}^{v,w}(u', u'', w', w'')\big(X_{t,u}^{v,w}\big)^2 \\
&+ \sum_{\substack{(t,u)\in D_1\\(v_1,w,v_2)\in P_2}} \sum_{(u',u'')\in I_{t,u}} s_{t,u}^{v_1 v_2}(u', u'')\big(X_{t,u}^{v_1,w} + X_{t,u}^{v_2,w}\big)^2 \\
&+ \sum_{\substack{(t_1,u,t_2)\in P_1\\(v,w)\in D_2}} \sum_{(w',w'')\in I_{v,w}} b_{t_1 t_2}^{v,w}(w', w'')\big(X_{t_1,u}^{v,w} + X_{t_2,u}^{v,w}\big)^2 \\
&+ \sum_{\substack{(t_1,u,t_2)\in P_1\\(v_1,w,v_2)\in P_2}} p_{t_1 t_2}^{v_1 v_2}\big(X_{t_1,u}^{v_1,w} + X_{t_1,u}^{v_2,w} + X_{t_2,u}^{v_1,w} + X_{t_2,u}^{v_2,w}\big)^2.
\end{aligned}$$

To prove Theorem 2, we express the terms involving one or two partial splits in terms of the X-variables. We use the following elementary linear-algebra identity.

Lemma 2 *Let*

$$A = \begin{pmatrix} a_1^1 & a_1^2 & a_1^3 \\ a_2^1 & a_2^2 & a_2^3 \end{pmatrix}$$

be a real 2×3 matrix. For $i, j \in \{1, 2, 3\}$, put

$$A^{i,j} := \begin{pmatrix} a_1^i & a_1^j \\ a_2^i & a_2^j \end{pmatrix}, \quad A^{1,2+3} := \begin{pmatrix} a_1^1 & a_1^2 + a_1^3 \\ a_2^1 & a_2^2 + a_2^3 \end{pmatrix},$$

and

$$A^{3,1+2} := \begin{pmatrix} a_1^3 & a_1^2 + a_1^1 \\ a_2^3 & a_2^2 + a_2^1 \end{pmatrix}.$$

Let $r_1 = a_1^1 + a_1^2 + a_1^3$ and $r_2 = a_2^1 + a_2^2 + a_2^3$, and assume $r_1, r_2 \neq 0$. For every $\alpha \in \mathbb{R}$,

$$\det(A^{1,3}) = \left(\alpha\frac{a_1^3}{r_1} + (1-\alpha)\frac{a_2^3}{r_2}\right)\det(A^{1,2+3}) - \left(\alpha\frac{a_1^1}{r_1} + (1-\alpha)\frac{a_2^1}{r_2}\right)\det(A^{3,1+2}).$$

Proof Append a copy of the first row of A above A to form a 3×3 matrix A_+. Its first two rows coincide, so $\det(A_+) = 0$. Expanding along the first row gives

$$0 = a_1^1 \det(A^{2,3}) - a_1^2 \det(A^{1,3}) + a_1^3 \det(A^{1,2}),$$

and hence

$$a_1^2 \det(A^{1,3}) = a_1^1 \det(A^{2,3}) + a_1^3 \det(A^{1,2}).$$

Adding $(a_1^1 + a_1^3)\det(A^{1,3})$ and dividing by r_1 proves the formula for $\alpha = 1$, because $\det(A^{1,2+3}) = \det(A^{1,2}) + \det(A^{1,3})$ and $\det(A^{3,1+2}) = -\det(A^{1,3}) - \det(A^{2,3})$. Appending a copy of the second row instead proves the case $\alpha = 0$. The general identity is the linear combination of these two cases with coefficients α and $1 - \alpha$. □

Proof of Theorem 2 We express the four contributions in Theorem 1 as quadratic functions of at most four variables $X_{t,u}^{v,w}$. For any $h \in \{1, 2\}$, write $\bar{h} := 3 - h$ for the complementary index. For edges tu of T_1 and vw of T_2,

$$\frac{1}{6} Q(S_{tu}, S_{vw}) = (X_{t,u}^{v,w})^2,$$

and this remains true after interchanging t, u and/or v, w. Directly from the definition of the coefficient family q and the row and column partitions,

$$\begin{aligned} 1 = &\sum_{\substack{u' \in N_{t+}(u),\ u'' \in N_+(u) \\ w' \in N_{v+}(w),\ w'' \in N_+(w)}} q_{t,u}^{v,w}(u', u'', w', w'') + \sum_{\substack{u' \in N_{t+}(u),\ u'' \in N_+(u) \\ v' \in N_{w+}(v),\ v'' \in N_+(v)}} q_{t,u}^{w,v}(u', u'', v', v'') \\ &+ \sum_{\substack{t' \in N_{u+}(t),\ t'' \in N_+(t) \\ w' \in N_{v+}(w),\ w'' \in N_+(w)}} q_{u,t}^{v,w}(t', t'', w', w'') + \sum_{\substack{t' \in N_{u+}(t),\ t'' \in N_+(t) \\ v' \in N_{w+}(v),\ v'' \in N_+(v)}} q_{u,t}^{w,v}(t', t'', v', v''). \end{aligned}$$

Associating each coefficient with the X-variable having the same outer directed-edge indices (t, u) and (v, w) gives

$$\frac{1}{6} \sum_{\substack{tu \in E_1 \\ vw \in E_2}} Q(S_{tu}, S_{vw}) = \sum_{\substack{(t,u) \in D_1 \\ (v,w) \in D_2}} \sum_{\substack{u' \in N_{t+}(u),\ u'' \in N_+(u) \\ w' \in N_{v+}(w),\ w'' \in N_+(w)}} q_{t,u}^{v,w}(u', u'', w', w'')(X_{t,u}^{v,w})^2. \quad (1)$$

For an edge tu of T_1 and a path $v_1 w v_2$ of length 2 in T_2,

$$\frac{1}{6} Q(S_{tu}, S_{v_1 v_2}) = \begin{vmatrix} a_{t,u}^{v_1,w}(t, v_2) & a_{t,u}^{v_2,w}(t, v_1) \\ a_{u,t}^{v_1,w}(u, v_2) & a_{u,t}^{v_2,w}(u, v_1) \end{vmatrix}^2.$$

Apply Lemma 2, with $\alpha = r_{t,u}(t)$, to

$$\begin{pmatrix} a_{t,u}^{v_1,w}(t, v_2) & r_{t,u}(t) - a_{t,u}^{v_1,w}(t, v_2) - a_{t,u}^{v_2,w}(t, v_1) & a_{t,u}^{v_2,w}(t, v_1) \\ a_{u,t}^{v_1,w}(u, v_2) & r_{u,t}(u) - a_{u,t}^{v_1,w}(u, v_2) - a_{u,t}^{v_2,w}(u, v_1) & a_{u,t}^{v_2,w}(u, v_1) \end{pmatrix}.$$

The two row sums in this application are the complementary positive branch masses $r_{t,u}(t)$ and $r_{u,t}(u) = 1 - r_{t,u}(t)$. Identifying the resulting 2×2 determinants with the corresponding X-variables gives

$$\frac{1}{6} Q(S_{tu}, S_{v_1 v_2}) = \left(c^{v_1,w}(v_2) X_{t,u}^{v_1,w} - c^{v_2,w}(v_1) X_{t,u}^{v_2,w}\right)^2. \quad (2)$$

Moreover,

$$c^{v_1,w}(v_2)^2 = \sum_{\substack{u' \in N_{t+}(u) \\ u'' \in N_+(u)}} q_{t,u}^{v_1,w}(u', u'', v_2, v_2) + \sum_{\substack{t' \in N_{u+}(t) \\ t'' \in N_+(t)}} q_{u,t}^{v_1,w}(t', t'', v_2, v_2),$$

and the corresponding identity holds with v_1 and v_2 interchanged. The mixed coefficient satisfies

$$c^{v_1,w}(v_2)c^{v_2,w}(v_1) = \sum_{\substack{u'\in N_{t+}(u)\\ u''\in N_+(u)}} s_{t,u}^{v_1v_2}(u',u'') + \sum_{\substack{t'\in N_{u+}(t)\\ t''\in N_+(t)}} s_{u,t}^{v_1v_2}(t',t'').$$

Expanding (2) gives two pure-square terms and one mixed term. The identities above give both pure-square coefficients, whereas P_2 contains only one orientation of each path: one pure-square term corresponds to the chosen orientation and the other to its reversal. Summing over all edges and path orbits therefore gives

$$\begin{aligned}\frac{1}{6}\sum_{\substack{tu\in E_1\\ v_1wv_2\in E_2^2}} Q(S_{tu},S_{v_1v_2}) = &\sum_{\substack{(t,u)\in D_1\\ (v_1,w,v_2)\in P_2}}\sum_{j\in\{1,2\}}\sum_{\substack{u'\in N_{t+}(u)\\ u''\in N_+(u)}} q_{t,u}^{v_j,w}(u',u'',v_{\bar{j}},v_{\bar{j}})(X_{t,u}^{v_j,w})^2\\ &-2\sum_{\substack{(t,u)\in D_1\\ (v_1,w,v_2)\in P_2}}\sum_{\substack{u'\in N_{t+}(u)\\ u''\in N_+(u)}} s_{t,u}^{v_1v_2}(u',u'')X_{t,u}^{v_1,w}X_{t,u}^{v_2,w}. \end{aligned} \tag{3}$$

For a path t_1ut_2 of length 2 in T_1 and an edge vw in T_2,

$$\frac{1}{6}Q(S_{t_1t_2},S_{vw}) = \begin{vmatrix} a_{t_1,u}^{v,w}(t_2,v) & a_{t_1,u}^{w,v}(t_2,w)\\ a_{t_2,u}^{v,w}(t_1,v) & a_{t_2,u}^{w,v}(t_1,w)\end{vmatrix}^2.$$

Apply Lemma 2 to

$$\begin{pmatrix} a_{t_1,u}^{v,w}(t_2,v) & c^{v,w}(v)-a_{t_1,u}^{v,w}(t_2,v)-a_{t_2,u}^{v,w}(t_1,v) & a_{t_2,u}^{v,w}(t_1,v)\\ a_{t_1,u}^{w,v}(t_2,w) & c^{w,v}(w)-a_{t_1,u}^{w,v}(t_2,w)-a_{t_2,u}^{w,v}(t_1,w) & a_{t_2,u}^{w,v}(t_1,w)\end{pmatrix}.$$

For $\alpha=1$ this yields

$$\begin{vmatrix} a_{t_1,u}^{v,w}(t_2,v) & a_{t_2,u}^{v,w}(t_1,v)\\ a_{t_1,u}^{w,v}(t_2,w) & a_{t_2,u}^{w,v}(t_1,w)\end{vmatrix} = \frac{a_{t_2,u}^{v,w}(t_1,v)}{c^{v,w}(v)}X_{t_2,u}^{v,w} - \frac{a_{t_1,u}^{v,w}(t_2,v)}{c^{v,w}(v)}X_{t_1,u}^{v,w},$$

whereas $\alpha=0$ gives

$$\begin{vmatrix} a_{t_1,u}^{v,w}(t_2,v) & a_{t_2,u}^{v,w}(t_1,v)\\ a_{t_1,u}^{w,v}(t_2,w) & a_{t_2,u}^{w,v}(t_1,w)\end{vmatrix} = \frac{a_{t_2,u}^{w,v}(t_1,w)}{c^{w,v}(w)}X_{t_2,u}^{v,w} - \frac{a_{t_1,u}^{w,v}(t_2,w)}{c^{w,v}(w)}X_{t_1,u}^{v,w}.$$

Multiplying these identities gives the square of the determinant on the left. Transposing its 2×2 matrix leaves the determinant unchanged and identifies its square with $Q(S_{t_1t_2},S_{vw})/6$. Changing $d=(v,w)$ to $d=(w,v)$ changes both displayed X-variables by a factor -1, so every quadratic monomial is unchanged. We therefore obtain the following formula for either $d=(d_1,d_2)\in\{(w,v),(v,w)\}$, where $X_{t_i,u}^{d} := X_{t_i,u}^{d_1,d_2}$:

$$\begin{aligned}\frac{1}{6}Q(S_{t_1t_2},S_{vw}) = \frac{1}{c^{v,w}(v)c^{w,v}(w)}\Big[&a_{t_2,u}^{v,w}(t_1,v)a_{t_2,u}^{w,v}(t_1,w)(X_{t_2,u}^{d})^2\\ &-\Big(a_{t_2,u}^{v,w}(t_1,v)a_{t_1,u}^{w,v}(t_2,w)+a_{t_1,u}^{v,w}(t_2,v)a_{t_2,u}^{w,v}(t_1,w)\Big)X_{t_2,u}^{d}X_{t_1,u}^{d}\\ &+a_{t_1,u}^{v,w}(t_2,v)a_{t_1,u}^{w,v}(t_2,w)(X_{t_1,u}^{d})^2\Big].\end{aligned}$$

Take the convex combination of the cases $d=(w,v)$ and $d=(v,w)$ with weights $c^{v,w}(v)$ and $c^{w,v}(w)$, respectively. These two weights are the normalized masses of the complementary sides of the edge split S_{vw}; hence they are nonnegative and

$$c^{v,w}(v)+c^{w,v}(w)=1.$$

Substitution of the coefficient definitions expresses this convex combination through the indexed q- and b-coefficients. Since P_1 contains one representative of each reversal pair, the sum below includes the pure-square terms associated with both endpoint orientations. Thus

$$\frac{1}{6}\sum_{\substack{t_1ut_2\in E_1^2\\ vw\in E_2}} Q(S_{t_1t_2},S_{vw}) = \sum_{\substack{(t_1,u,t_2)\in P_1\\ (v,w)\in D_2}}\sum_{i\in\{1,2\}}\sum_{\substack{w'\in N_{v+}(w)\\ w''\in N_+(w)}} q^{v,w}_{t_i,u}(t_{\bar{i}},t_{\bar{i}},w',w'')(X^{v,w}_{t_i,u})^2$$
$$-2\sum_{\substack{(t_1,u,t_2)\in P_1\\ (v,w)\in D_2}}\sum_{\substack{w'\in N_{v+}(w)\\ w''\in N_+(w)}} b^{v,w}_{t_1t_2}(w',w'')X^{v,w}_{t_2,u}X^{v,w}_{t_1,u}. \tag{4}$$

If t_1ut_2 and v_1wv_2 are paths of length 2 in T_1 and T_2, respectively, then

$$\frac{1}{6}Q(S_{t_1t_2},S_{v_1v_2}) = \begin{vmatrix} a^{v_1,w}_{t_1,u}(t_2,v_2) & a^{v_2,w}_{t_1,u}(t_2,v_1)\\ a^{v_1,w}_{t_2,u}(t_1,v_2) & a^{v_2,w}_{t_2,u}(t_1,v_1)\end{vmatrix}^2.$$

Apply Lemma 2 to

$$\begin{pmatrix} a^{v_1,w}_{t_1,u}(t_2,v_2) & c^{v_1,w}(v_2)-a^{v_1,w}_{t_1,u}(t_2,v_2)-a^{v_1,w}_{t_2,u}(t_1,v_2) & a^{v_1,w}_{t_2,u}(t_1,v_2)\\ a^{v_2,w}_{t_1,u}(t_2,v_1) & c^{v_2,w}(v_1)-a^{v_2,w}_{t_1,u}(t_2,v_1)-a^{v_2,w}_{t_2,u}(t_1,v_1) & a^{v_2,w}_{t_2,u}(t_1,v_1)\end{pmatrix}.$$

For $\alpha=1$, repeated use of the determinant identity used to derive (2) gives

$$\begin{vmatrix} a^{v_1,w}_{t_1,u}(t_2,v_2) & a^{v_1,w}_{t_2,u}(t_1,v_2)\\ a^{v_2,w}_{t_1,u}(t_2,v_1) & a^{v_2,w}_{t_2,u}(t_1,v_1)\end{vmatrix} = \frac{a^{v_1,w}_{t_2,u}(t_1,v_2)}{c^{v_1,w}(v_2)}\Big(c^{v_2,w}(v_1)X^{v_2,w}_{t_2,u}-c^{v_1,w}(v_2)X^{v_1,w}_{t_2,u}\Big)$$
$$-\frac{a^{v_1,w}_{t_1,u}(t_2,v_2)}{c^{v_1,w}(v_2)}\Big(c^{v_2,w}(v_1)X^{v_2,w}_{t_1,u}-c^{v_1,w}(v_2)X^{v_1,w}_{t_1,u}\Big).$$

Correspondingly, $\alpha=0$ yields

$$\begin{vmatrix} a^{v_1,w}_{t_1,u}(t_2,v_2) & a^{v_1,w}_{t_2,u}(t_1,v_2)\\ a^{v_2,w}_{t_1,u}(t_2,v_1) & a^{v_2,w}_{t_2,u}(t_1,v_1)\end{vmatrix} = \frac{a^{v_2,w}_{t_2,u}(t_1,v_1)}{c^{v_2,w}(v_1)}\Big(c^{v_2,w}(v_1)X^{v_2,w}_{t_2,u}-c^{v_1,w}(v_2)X^{v_1,w}_{t_2,u}\Big)$$
$$-\frac{a^{v_2,w}_{t_1,u}(t_2,v_1)}{c^{v_2,w}(v_1)}\Big(c^{v_2,w}(v_1)X^{v_2,w}_{t_1,u}-c^{v_1,w}(v_2)X^{v_1,w}_{t_1,u}\Big).$$

Multiplying the two equations gives the square of the determinant on the left. Transposition leaves this determinant unchanged and identifies its square with $Q(S_{t_1t_2},S_{v_1v_2})/6$. Substituting the definitions of the indexed q-, b-, s-, and p-coefficients gives

$$\frac{1}{6}Q(S_{t_1t_2},S_{v_1v_2}) = \sum_{i,j\in\{1,2\}} q^{v_j,w}_{t_i,u}(t_{\bar{i}},t_{\bar{i}},v_{\bar{j}},v_{\bar{j}})(X^{v_j,w}_{t_i,u})^2$$
$$-2\sum_{j\in\{1,2\}} b^{v_j,w}_{t_1t_2}(v_{\bar{j}},v_{\bar{j}})X^{v_j,w}_{t_1,u}X^{v_j,w}_{t_2,u}$$
$$-2\sum_{i\in\{1,2\}} s^{v_1v_2}_{t_i,u}(t_{\bar{i}},t_{\bar{i}})X^{v_1,w}_{t_i,u}X^{v_2,w}_{t_i,u}$$
$$+2p^{v_1v_2}_{t_1t_2}\Big(X^{v_1,w}_{t_1,u}X^{v_2,w}_{t_2,u}+X^{v_2,w}_{t_1,u}X^{v_1,w}_{t_2,u}\Big). \tag{5}$$

Set

$$K_{t,u} := N_{t+}(u)\times N_+(u), \qquad \Delta_{t,u} := \{(u',u') : u'\in N_t(u)\}.$$

Then $K_{t,u}\setminus\Delta_{t,u} = I'_{t,u}$. For the pure-square terms whose coefficients belong to the q-family, inclusion–exclusion gives

$$\mathbf{1}_{K_{t,u}\times K_{v,w}} - \mathbf{1}_{K_{t,u}\times\Delta_{v,w}} - \mathbf{1}_{\Delta_{t,u}\times K_{v,w}} + \mathbf{1}_{\Delta_{t,u}\times\Delta_{v,w}} = \mathbf{1}_{I'_{t,u}\times I'_{v,w}}.$$

For the s-mixed terms, the corresponding row-coordinate identity is $\mathbf{1}_{K_{t,u}} - \mathbf{1}_{\Delta_{t,u}} = \mathbf{1}_{I'_{t,u}}$; for the b-mixed terms, the column-coordinate identity is the analogous formula with (v,w) in place of (t,u). The mixed terms carrying a p-coefficient occur only in the path–path contribution. Substituting (1), (3), (4), and (5) into Theorem 1 and collecting like monomials gives

$$\begin{aligned}\frac{1}{6}Q(\mathcal{T}_1,\mathcal{T}_2) = &\sum_{\substack{(t,u)\in D_1\\(v,w)\in D_2}}\ \sum_{\substack{(u',u'')\in I'_{t,u}\\(w',w'')\in I'_{v,w}}} q_{t,u}^{v,w}(u',u'',w',w'')(X_{t,u}^{v,w})^2\\ &+2\sum_{\substack{(t,u)\in D_1\\(v_1,w,v_2)\in P_2}}\ \sum_{(u',u'')\in I'_{t,u}} s_{t,u}^{v_1v_2}(u',u'')X_{t,u}^{v_1,w}X_{t,u}^{v_2,w}\\ &+2\sum_{\substack{(t_1,u,t_2)\in P_1\\(v,w)\in D_2}}\ \sum_{(w',w'')\in I'_{v,w}} b_{t_1t_2}^{v,w}(w',w'')X_{t_1,u}^{v,w}X_{t_2,u}^{v,w}\\ &+2\sum_{\substack{(t_1,u,t_2)\in P_1\\(v_1,w,v_2)\in P_2}} p_{t_1t_2}^{v_1v_2}\Big(X_{t_1,u}^{v_1,w}X_{t_2,u}^{v_2,w}+X_{t_1,u}^{v_2,w}X_{t_2,u}^{v_1,w}\Big).\end{aligned}\tag{6}$$

We next compare this quadratic polynomial with the expansion of the right-hand side of Theorem 2. For each directed edge put

$$B_{t,u} := I'_{t,u}\setminus I_{t,u} = \{(u',t) : u'\in N_t(u)\},\qquad I'_{t,u} = I_{t,u}\,\dot\cup\, B_{t,u},$$

and define $B_{v,w}$ analogously. The definitions of b, s, and p give the following indexed boundary identities:

$$\begin{aligned}q_{t,u}^{v,w}(u',u'',w',v) &= s_{t,u}^{vw'}(u',u'') &&\text{if } (w',v)\in B_{v,w},\\ q_{t,u}^{v,w}(u',t,w',w'') &= b_{tu'}^{v,w}(w',w'') &&\text{if } (u',t)\in B_{t,u},\\ q_{t,u}^{v,w}(u',t,w',v) &= p_{tu'}^{vw'} &&\text{if } (u',t)\in B_{t,u} \text{ and } (w',v)\in B_{v,w},\\ s_{t,u}^{vw'}(u',t) &= p_{tu'}^{vw'} &&\text{if } (u',t)\in B_{t,u} \text{ and } (w',v)\in B_{v,w},\\ b_{tu'}^{v,w}(w',v) &= p_{tu'}^{vw'} &&\text{if } (u',t)\in B_{t,u} \text{ and } (w',v)\in B_{v,w}.\end{aligned}$$

These equalities concern only the coefficients written above at the stated indices; they do not assert equality of the complete q, s, b, or p coefficient families. The disjoint decomposition

$$I'_{t,u}\times I'_{v,w} = (I_{t,u}\times I_{v,w})\,\dot\cup\,(I_{t,u}\times B_{v,w})\,\dot\cup\,(B_{t,u}\times I_{v,w})\,\dot\cup\,(B_{t,u}\times B_{v,w})$$

and the preceding identities give the following correspondence. In the first factor of each Cartesian product, I and B mean membership in $I_{t,u}$ and $B_{t,u}$; in the second factor, they mean membership in $I_{v,w}$ and $B_{v,w}$. Within an edge–path term write $X_j := X_{t,u}^{v_j,w}$; within a path–edge term write $X_i := X_{t_i,u}^{v,w}$; and within a path–path term write $X_{ij} := X_{t_i,u}^{v_j,w}$. Thus $i\in\{1,2\}$ selects the endpoint t_i, equivalently the directed edge (t_i,u), of the fixed first-tree path, and $j\in\{1,2\}$ selects v_j, equivalently (v_j,w), of the fixed second-tree path. In an s-row of the table, a lone I or B records membership of the first-tree pair (u',u'') in $I_{t,u}$ or $B_{t,u}$; in a b-row it records membership of the second-tree pair (w',w'') in $I_{v,w}$ or $B_{v,w}$. The bare letters q,s,b,p in the table below are abbreviations for the previously defined coefficient families carrying the same letters,

$$q_{t,u}^{v,w}(u',u'',w',w''),\qquad s_{t,u}^{v_1v_2}(u',u''),\qquad b_{t_1t_2}^{v,w}(w',w''),\qquad p_{t_1t_2}^{v_1v_2};$$

they are not new quantities. Indices fixed by the row of the table and by the surrounding sums are suppressed only to keep the table readable. If λ denotes the relevant coefficient, a

pure-square term means λX_α^2, whereas a *mixed term* means $2\lambda X_\alpha X_\beta$ with $\alpha \neq \beta$. For the p-family, view the variables as the 2×2 array (X_{ij}). Thus an s-mixed term joins entries in the same row of this array, a b-mixed term joins entries in the same column, and the explicit p-term in (6) consists of the two opposite-corner mixed terms:

term in (6)	corresponding term in Theorem 2
$qX^2, I \times I$	pure-square term in the q-sum
$qX^2, I \times B$	pure-square term in the s-sum (first boundary identity)
$qX^2, B \times I$	pure-square term in the b-sum (second boundary identity)
$qX^2, B \times B$	pure-square term in the p-sum (third boundary identity)
$2sX_jX_{\bar{j}}, I$	mixed term in the s-sum
$2sX_jX_{\bar{j}}, B$	same-row mixed term in the p-sum (fourth identity)
$2bX_iX_{\bar{i}}, I$	mixed term in the b-sum
$2bX_iX_{\bar{i}}, B$	same-column mixed term in the p-sum (fifth identity)
$2p(X_{11}X_{22} + X_{12}X_{21})$	opposite-corner mixed terms in the p-sum

The four Cartesian products in the displayed decomposition are disjoint. Moreover, the derivations of (3), (4), and (5) already combine the contributions from both orientations of each reversal orbit, whereas P_1 and P_2 select one representative of each orbit. Hence every indexed term in (6) occurs once in the expansion of the four sums of squares. The two expressions are equal, which proves the theorem. □

Lemma 3 (Partial-quartet reconstruction) *Let $\mathcal{T}$ be an X-tree and let $A \mid (X \setminus A)$ be a full split, so $A \neq \varnothing$ and $X \setminus A \neq \varnothing$. We say that an edge e represents this split when the unordered split S_e equals $A \mid (X \setminus A)$. The split is represented by an edge of $\mathcal{T}$ if and only if every partial quartet displayed by it is displayed by $\mathcal{T}$. Consequently the three systems $Q_2(\mathcal{T})$, $Q_3(\mathcal{T})$, and $Q_4(\mathcal{T})$ determine $\mathcal{T}$ up to a label-preserving graph isomorphism.*

Moreover, if $|X| \geq 4$, an X-tree that displays $xx \mid yz$ for every three distinct taxa x, y, z and displays a full quartet on every four-set is binary phylogenetic.

Proof Only the reverse implication in the first assertion needs proof. Choose $b_2 \in X \setminus A$ and root the graph at its labelled vertex $\phi(b_2)$. For a vertex, its rooted cluster is the union of the label fibres in its descendant subtree. Among the rooted clusters containing A, choose one of minimum cardinality, say C with root vertex v. If $C = A$, the edge above v represents the proposed split. If $C \neq A$, minimality supplies two possibly equal taxa $a_1, a_2 \in A$ such that the path between $\phi(a_1)$ and $\phi(a_2)$ contains v: either v itself carries a taxon of A, or A meets at least two child branches of v, since otherwise a smaller child cluster would contain A. Choose $b_1 \in C \setminus A$; the path from $\phi(b_2)$ to $\phi(b_1)$ also contains v. The partial quartet $a_1a_2 \mid b_1b_2$ is displayed by $A \mid (X \setminus A)$, but no tree edge can display it: such an edge would put v simultaneously on both sides of the cut, because v lies on both within-side paths. This contradiction proves the criterion.

If two trees have the same three partial-quartet systems, apply the criterion first to every edge split of the first tree using the second tree, and then vice versa, to obtain the same full edge-split system. After rooting each graph at the labelled vertex of the same taxon, the sides away from the root form the same laminar family of rooted clusters. Inclusion in this family recovers the parent–child relation, and the label fibre at a vertex is its cluster minus the disjoint union of its child clusters. This reconstructs the required label-preserving isomorphism.

For the last assertion, the three-taxon condition first forces distinct taxa to occupy distinct vertices. If a labelled vertex were not a leaf, choose taxa in two different branches at that vertex; then the required split $xx \mid yz$ could not be displayed. Hence every taxon labels a distinct leaf and every leaf is labelled; moreover, there are no degree-2 vertices. Finally a vertex of degree at least 4 has four labelled branches; choosing one taxon in each produces a four-set with no displayed 2–2 split. Thus every nonleaf vertex has degree 3. □

Lemma 4 (Vertex–side independence bridge) *Let U be uniform on X and put $Z_i = \phi_i(U)$ for $i \in \{1,2\}$. The trees $\mathcal{T}_1, \mathcal{T}_2$ are independent if and only if the finite-valued random variables Z_1, Z_2 are independent. If they are not independent, there exist vertices $u \in V_1$, $w \in V_2$ with $P(Z_1 = u) > 0$ and $P(Z_2 = w) > 0$, and incident directed edges $(t,u) \in D_1$, $(v,w) \in D_2$ such that*

$$Q(S_{tu}, S_{vw}) > 0, \qquad r_{t,u}(u) > 0, \qquad c^{v,w}(w) > 0.$$

Proof Every side of an edge split is a union of vertex events, so independence of Z_1, Z_2 implies independence of all edge-split pairs. Conversely, root each tree. For a vertex z of tree i, let $C_z^{(i)}$ be the union of the label fibres in the descendant subtree rooted at z, with $C_z^{(i)} = X$ at the root. For $C \subseteq X$, write $\mathbf{1}_C := \mathbf{1}_{\{U \in C\}}$. If y ranges over the children of z, then

$$\mathbf{1}_{\{Z_i = z\}} = \mathbf{1}_{C_z^{(i)}} - \sum_{y \text{ child of } z} \mathbf{1}_{C_y^{(i)}}.$$

Every non-root cluster is a side of an edge split. If all edge-split pairs are independent, every cross-covariance between their side indicators is zero. Bilinearity of covariance therefore makes every covariance between a Z_1-singleton indicator and a Z_2-singleton indicator zero. Since these are Bernoulli indicators, this says

$$P(Z_1 = z, Z_2 = z') = P(Z_1 = z)P(Z_2 = z')$$

for every pair of vertices z, z', which is exactly independence of Z_1, Z_2.

Now suppose the trees are not independent. Then some vertices u, w with positive marginal probabilities satisfy $\mathrm{Cov}(\mathbf{1}_{\{Z_1=u\}}, \mathbf{1}_{\{Z_2=w\}}) \neq 0$. Root the trees at u, w. Neither random variable is constant, so both roots have incident edges, and

$$\mathbf{1}_{\{Z_1=u\}} = 1 - \sum_{t \in N(u)} \mathbf{1}_{C_t^{(1)}}, \qquad \mathbf{1}_{\{Z_2=w\}} = 1 - \sum_{v \in N(w)} \mathbf{1}_{C_v^{(2)}}.$$

Bilinearity supplies $t \in N(u)$ and $v \in N(w)$ for which the two branch indicators have nonzero covariance. Their 2×2 intersection table has nonzero determinant, hence $Q(S_{tu}, S_{vw}) > 0$. Orienting both edges toward the roots gives $r_{t,u}(u) = P(Z_1 = u) > 0$ and $c^{v,w}(w) = P(Z_2 = w) > 0$. □

Corollary 1 *For two X-trees $\mathcal{T}_1$ and $\mathcal{T}_2$, with $n = |X|$,*

$$0 \leq Q(\mathcal{T}_1, \mathcal{T}_2) \leq 1 - \frac{4n-3}{n^3}.$$

The lower bound is attained if and only if $\mathcal{T}_1$ and $\mathcal{T}_2$ are independent. If $n \geq 4$, then the upper bound is attained if and only if the trees are binary phylogenetic and are label-preservingly isomorphic.

Proof The lower bound follows from Theorem 2, and the quartet covariance between independent X-trees is zero. Conversely, Lemma 4 supplies an edge-split pair with positive quartet covariance for which $r_{t,u}(u) > 0$ and $c^{v,w}(w) > 0$. Lemma 6 below then gives

$$Q(\mathcal{T}_1, \mathcal{T}_2) \geq \tfrac{1}{2} c^{v,w}(w) r_{t,u}(u) Q(S_{tu}, S_{vw}) > 0.$$

The probability that some taxon is drawn at least three times among the four draws is $(4n-3)/n^3$, and every such ordered quadruple contributes zero to $Q(\mathcal{T}_1, \mathcal{T}_2)$. Every other ordered quadruple contributes at most 1, proving the upper bound. Equality means that every ordered quadruple outside the event just excluded contributes 1: the two trees display the same partial quartet on every support of size 2 or 3, and the same full quartet on every four-set. If $n \geq 4$, the final assertion of Lemma 3 makes both trees binary phylogenetic, and its reconstruction assertion makes them label-preservingly isomorphic. Conversely, a binary phylogenetic tree displays every non-full partial quartet and exactly one full quartet on each four-set. Two label-preservingly isomorphic such trees therefore make every eligible ordered quadruple contribute 1, so equality holds. □

To relate quartet covariance to quartet distance, we record the following counting identity.

Lemma 5 *For two X-trees $\mathcal{T}_1$ and $\mathcal{T}_2$,*

$$Q(\mathcal{T}_1, \mathcal{T}_2) = \frac{1}{|X|^4}\Big(24|Q_4(\mathcal{T}_1) \cap Q_4(\mathcal{T}_2)| + 12|Q_3(\mathcal{T}_1) \cap Q_3(\mathcal{T}_2)| + 6|Q_2(\mathcal{T}_1) \cap Q_2(\mathcal{T}_2)| - 12 d_Q(\mathcal{T}_1, \mathcal{T}_2)\Big).$$

Proof For a partial quartet $x_1x_1 \mid x_2x_2$, there are 6 ordered 4-tuples containing x_1 and x_2 twice. For $x_1x_1 \mid x_2x_3$, there are 12 ordered 4-tuples containing x_1 twice and x_2, x_3 once. For a 4-set there are 24 ordered 4-tuples containing each element once. A common displayed object contributes $+1$ for each ordering, whereas a four-set counted by d_Q contributes $-1/2$ for each of its 24 orderings. Every ordered quadruple has probability $|X|^{-4}$, which proves the formula. □

The 2/3-conjecture is now a direct consequence of Theorem 2.

Corollary 2 *For two binary phylogenetic X-trees $\mathcal{T}_1$ and $\mathcal{T}_2$,*

$$d_Q(\mathcal{T}_1, \mathcal{T}_2) \leq \frac{2}{3}\binom{|X|}{4} + \binom{|X|}{3} + \frac{1}{6}\binom{|X|}{2}.$$

Proof By Lemma 5,

$$Q(\mathcal{T}_1, \mathcal{T}_2) = \frac{1}{|X|^4}\left(24\left(\binom{|X|}{4} - d_Q(\mathcal{T}_1, \mathcal{T}_2)\right) - 12 d_Q(\mathcal{T}_1, \mathcal{T}_2) + 36\binom{|X|}{3} + 6\binom{|X|}{2}\right),$$

because every 4-set induces one quartet in each tree, while all non-full partial quartets are displayed by both binary phylogenetic trees. The nonnegativity of Q yields the claimed inequality. □

Pachter's 2026 preprint (arXiv:2608.03542) also proves the same asymptotic 2/3 conclusion by a common-root planarisation argument and a five-leaf identity. Our proof instead derives the displayed finite-$|X|$ inequality from the nonnegative sum-of-squares representation in Theorem 2.

For $|X| \geq 2$ this explicit bound is not sharp, because two phylogenetic trees are never independent. For any taxon x, compare the two pendant edge splits having singleton side $\{x\}$. Their 2×2 intersection table has determinant $|X| - 1 \neq 0$, so their split covariance is positive. On the other hand, the difference between the upper bound and the infinite family of tree pairs from [14] is only $O(|X|^2 \log |X|)$.

We end this section with a lower bound on the coefficient multiplying $(X_{t,u}^{v,w})^2$ in the q-square family, namely the first nonnegative sum in Theorem 2.

Lemma 6 *For two X-trees $\mathcal{T}_1$ and $\mathcal{T}_2$, and directed edges $(t,u) \in D_1$, $(v,w) \in D_2$,*

$$\sum_{\substack{(u',u'') \in I_{t,u} \\ (w',w'') \in I_{v,w}}} q_{t,u}^{v,w}(u',u'',w',w'') \geq \frac{1}{2} c^{v,w}(w) r_{t,u}(u).$$

In particular,

$$Q(\mathcal{T}_1, \mathcal{T}_2) \geq \frac{1}{2} \sum_{(t,u) \in D_1} \sum_{(v,w) \in D_2} c^{v,w}(w) r_{t,u}(u) Q(S_{tu}, S_{vw}).$$

Proof Fix $(t,u) \in D_1$ and $(v,w) \in D_2$, and abbreviate

$$a(x,y) := a_{t,u}^{v,w}(x,y), \qquad c(y) := c^{v,w}(y), \qquad r(x) := r_{t,u}(x).$$

For $w' \in N_{v+}(w)$, put

$$J(w') := \{w'' \in N_+(w) : w' = w \text{ or } w'' \notin \{v, w'\}\}, \qquad \sigma(w') := \sum_{w'' \in J(w')} c(w'').$$

The pair (w', w) belongs to $I_{v,w}$ for every $w' \in N_{v+}(w)$, so $\sigma(w') \geq c(w)$. If $w' = w$, then every $w'' \in N_+(w)$ is allowed, whence $J(w) = N_+(w)$ and $\sigma(w) = 1$.

All q-coefficients are nonnegative. Since $u' = u$ makes every $u'' \in N_+(u)$ admissible in $I_{t,u}$, retaining only those terms gives

$$\sum_{(u',u'') \in I_{t,u}} \sum_{(w',w'') \in I_{v,w}} q_{t,u}^{v,w}(u',u'',w',w'') \geq \sum_{w' \in N_{v+}(w)} \sum_{w'' \in J(w')} \sum_{u'' \in N_+(u)} q_{t,u}^{v,w}(u,u'',w',w'').$$

The row blocks indexed by $N_+(u)$ partition X, so

$$\sum_{u'' \in N_+(u)} a(u'', v) = c(v), \qquad \sum_{u'' \in N_+(u)} a(u'', w') = c(w').$$

The definition of the coefficient family q turns the retained triple sum into

$$\frac{1}{2} \sum_{w' \in N_{v+}(w)} \sigma(w') \left(a(u,w') + c(w') \frac{a(u,v)}{c(v)} \right). \tag{7}$$

In T_2, the component of $T_2 - vw$ containing v contains at least one label, so $0 < c(v) \leq 1$. The column blocks indexed by $N_{v+}(w)$ are all column blocks except that indexed by v. Hence

$$\sum_{w' \in N_{v+}(w)} \sigma(w') a(u,w') \geq c(w)\big(r(u) - a(u,v)\big),$$

$$\sum_{w' \in N_{v+}(w)} \sigma(w')c(w')\frac{a(u,v)}{c(v)} \geq \sigma(w)c(w)\frac{a(u,v)}{c(v)} \geq c(w)a(u,v).$$

Substitution into (7) proves the first claim.

For the final assertion, retain only the q-square family, namely the first nonnegative sum in Theorem 2, and use $Q(S_{tu}, S_{vw}) = 6(X^{v,w}_{t,u})^2$:

$$\begin{aligned}\frac{1}{6}Q(\mathcal{T}_1, \mathcal{T}_2) &\geq \sum_{(t,u)\in D_1}\sum_{(v,w)\in D_2}\left(\sum_{(u',u'')\in I_{t,u}}\sum_{(w',w'')\in I_{v,w}} q^{v,w}_{t,u}(u',u'',w',w'')\right)(X^{v,w}_{t,u})^2 \\ &\geq \frac{1}{12}\sum_{(t,u)\in D_1}\sum_{(v,w)\in D_2} c^{v,w}(w)r_{t,u}(u)Q(S_{tu}, S_{vw}).\end{aligned}$$

Multiplication by 6 proves the stated bound. □

4 Applications to dependence measures

4.1 Normalised quartet dependence measures

The quartet covariance is a similarity measure for X-trees related to the quartet distance. Nevertheless, quartet covariance never reaches 1, even when the compared trees are identical. A low value may reflect incompatibility or lack of resolution. We therefore consider two normalisations.

For two non-trivial X-trees, define the *quartet compatibility index* $c_Q(\mathcal{T}_1, \mathcal{T}_2)$ to be the quartet covariance divided by the sum of (i) the probability of drawing a partial quartet displayed by both trees and (ii) half the probability that the draw induces distinct full quartets in the two trees. Its denominator can be computed from Theorem 1 by replacing the determinant in $Q(S_1, S_2)$ with the permanent of the same matrix. Here

$$\operatorname{per}\begin{pmatrix} a & b \\ c & d \end{pmatrix} := ad + bc,$$

in contrast to the determinant $ad - bc$. Equivalently,

$$c_Q(\mathcal{T}_1, \mathcal{T}_2) = \frac{24|Q_4(\mathcal{T}_1) \cap Q_4(\mathcal{T}_2)| + 12|Q_3(\mathcal{T}_1) \cap Q_3(\mathcal{T}_2)| + 6|Q_2(\mathcal{T}_1) \cap Q_2(\mathcal{T}_2)| - 12d_Q(\mathcal{T}_1, \mathcal{T}_2)}{24|Q_4(\mathcal{T}_1) \cap Q_4(\mathcal{T}_2)| + 12|Q_3(\mathcal{T}_1) \cap Q_3(\mathcal{T}_2)| + 6|Q_2(\mathcal{T}_1) \cap Q_2(\mathcal{T}_2)| + 12d_Q(\mathcal{T}_1, \mathcal{T}_2)}.$$

This denominator is strictly positive. Indeed, choose one edge split $A \mid B$ from $\mathcal{T}_1$ and one edge split $C \mid D$ from $\mathcal{T}_2$. Since all four sides are nonempty, either both $A \cap C$ and $B \cap D$ are nonempty, or both $A \cap D$ and $B \cap C$ are nonempty. Choosing one taxon from each of the corresponding two intersections gives a partial quartet in $Q_2(\mathcal{T}_1) \cap Q_2(\mathcal{T}_2)$.

Two full splits $A \mid B$ and $C \mid D$ are called *compatible* if at least one of $A \cap C$, $A \cap D$, $B \cap C$, and $B \cap D$ is empty. A split system is compatible if every pair of its splits is compatible.

Proposition 1 *For two non-trivial X-trees $\mathcal{T}_1$ and $\mathcal{T}_2$, $0 \leq c_Q(\mathcal{T}_1, \mathcal{T}_2) \leq 1$. Moreover, $c_Q(\mathcal{T}_1, \mathcal{T}_2) = 0$ if and only if the trees are independent. Furthermore, $c_Q(\mathcal{T}_1, \mathcal{T}_2) = 1$ if and only if the union of their edge-split systems is compatible.*

Proof By the preceding definition, compatible splits cannot display conflicting partial quartets. Conversely, if two edge splits are incompatible, choosing one taxon from each of their four nonempty cross-intersections produces a conflicting quartet. Hence the union of the two edge-split systems is compatible exactly when $d_Q(\mathcal{T}_1, \mathcal{T}_2) = 0$. The defining fraction for c_Q is equal to 1 exactly when this conflicting term vanishes. The zero statement follows from Corollary 1 and the strict positivity of the denominator established above. □

To measure similarity rather than compatibility, define the *quartet correlation*

$$\rho_Q(\mathcal{T}_1, \mathcal{T}_2) := \frac{Q(\mathcal{T}_1, \mathcal{T}_2)}{\sqrt{Q(\mathcal{T}_1, \mathcal{T}_1)Q(\mathcal{T}_2, \mathcal{T}_2)}}.$$

Proposition 2 *For two non-trivial X-trees $\mathcal{T}_1$ and $\mathcal{T}_2$, $0 \le \rho_Q(\mathcal{T}_1, \mathcal{T}_2) \le c_Q(\mathcal{T}_1, \mathcal{T}_2)$. Furthermore, $\rho_Q = 0$ if and only if the trees are independent, while $0 < \rho_Q = c_Q$ if and only if the trees are label-preservingly isomorphic and $Q(\mathcal{T}_1, \mathcal{T}_1) > 0$. In the latter case, $\rho_Q = c_Q = 1$.*

Proof For $i \in \{1, 2\}$,

$$Q(\mathcal{T}_i, \mathcal{T}_i) = \frac{1}{|X|^4}\left(24|Q_4(\mathcal{T}_i)| + 12|Q_3(\mathcal{T}_i)| + 6|Q_2(\mathcal{T}_i)|\right).$$

Each self-comparison contains every common displayed partial quartet. For each $i \in \{1, 2\}$, the full-quartet system $Q_4(\mathcal{T}_i)$ also contains one quartet for each of the d_Q four-sets on which the two trees disagree. Thus, in particular,

$$|Q_4(\mathcal{T}_i)| \ge |Q_4(\mathcal{T}_1) \cap Q_4(\mathcal{T}_2)| + d_Q(\mathcal{T}_1, \mathcal{T}_2),$$

with the analogous common-system containments for Q_2 and Q_3. Because each tree is nontrivial, it displays at least one support-2 partial quartet; hence both self covariances are positive. Comparing the preceding self sums with the numerator and denominator of c_Q gives, for $i \in \{1, 2\}$,

$$\frac{Q(\mathcal{T}_1, \mathcal{T}_2)}{Q(\mathcal{T}_i, \mathcal{T}_i)} \le c_Q(\mathcal{T}_1, \mathcal{T}_2).$$

Multiplying the two inequalities and taking nonnegative square roots gives $\rho_Q \le c_Q$. Moreover, $\rho_Q = 0$ is now equivalent to $Q(\mathcal{T}_1, \mathcal{T}_2) = 0$, and hence to independence by Corollary 1.

Suppose now that $0 < \rho_Q = c_Q$. Then the two inequalities above are both equalities and their common numerator is positive. Put

$$A := 24|Q_4(\mathcal{T}_1) \cap Q_4(\mathcal{T}_2)| + 12|Q_3(\mathcal{T}_1) \cap Q_3(\mathcal{T}_2)| + 6|Q_2(\mathcal{T}_1) \cap Q_2(\mathcal{T}_2)|,$$
$$B := 12d_Q(\mathcal{T}_1, \mathcal{T}_2).$$

Thus A is the weighted number of common displayed objects. The denominator of c_Q is $A + B$, whereas, for $i \in \{1, 2\}$, the weighted self-comparison count $|X|^4 Q(\mathcal{T}_i, \mathcal{T}_i)$ is $A + 2B$ plus the nonnegative weighted count of the remaining partial quartets displayed only by that tree, excluding the disagreement quartets already included in $2B$. Equality of the positive ratios therefore forces $B = 0$ and both exclusive counts to vanish. Thus the two trees have the same displayed partial quartets of support 2, 3, and 4. Those systems determine the X-tree up to a graph isomorphism preserving every taxon label by Lemma 3. Conversely, a label-preserving isomorphism makes the two displayed systems and both self covariances equal; if the self covariance is positive, both normalized indices are 1. This proves the stated equality characterization. □

4.2 Tree-metric distance covariance

One can also weight partial quartets by their strength of support. This is especially useful for trees equipped with edge lengths. An *edge-weighted X-tree* is a triple $\mathcal{T} = (T, \phi, l)$, where (T, ϕ) is an X-tree and $l : E(T) \to (0, \infty)$ assigns a positive length to each edge. An unweighted tree is recovered by taking $l(e) = 1$ for all e. The *length* of a partial quartet $x_1x_2 \mid x_3x_4$ is the sum of the lengths of all edges uv for which S_{uv} displays it. The tree also defines a pseudometric $d_{\mathcal{T}}$ on X: for distinct taxa x_1, x_2, $d_{\mathcal{T}}(x_1, x_2)$ is the length of $x_1x_1 \mid x_2x_2$, and $d_{\mathcal{T}}(x, x) := 0$.

For two edge-weighted X-trees $\mathcal{T}_i = (T_i, \phi_i, l_i)$, define their *distance covariance* $\mathrm{dCov}^2(\mathcal{T}_1, \mathcal{T}_2)$ by multiplying every ordered quadruple contribution to $Q(\mathcal{T}_1, \mathcal{T}_2)$ by the lengths of its two induced partial quartets. More explicitly,

$$\mathrm{dCov}^2(\mathcal{T}_1, \mathcal{T}_2) := \sum_{\substack{tu \in E_1 \\ vw \in E_2}} Q(S_{tu}, S_{vw}) l_1(tu) l_2(vw).$$

This quantity is nonnegative, and it is zero exactly when the underlying X-trees are independent. Define their *distance correlation* by

$$\mathrm{dCor}(\mathcal{T}_1, \mathcal{T}_2) := \frac{\mathrm{dCov}^2(\mathcal{T}_1, \mathcal{T}_2)}{\sqrt{\mathrm{dCov}^2(\mathcal{T}_1, \mathcal{T}_1)\, \mathrm{dCov}^2(\mathcal{T}_2, \mathcal{T}_2)}}.$$

When the denominator is zero, set $\mathrm{dCor}(\mathcal{T}_1, \mathcal{T}_2) := 0$. This ratio is the quantity often denoted by squared distance correlation; throughout this paper we use the shorter symbol dCor for that convention. These names are standard in statistics for measures of dependence between random vectors; we now justify the same notation for edge-weighted X-trees.

Székely et al. [15] introduced distance covariance in 2007 to detect nonlinear dependence between Euclidean random variables. For an n-point sample $\{(x_i, y_i) : 1 \le i \le n\}$, empirical distance covariance is defined from the pairwise distances in the two coordinates. Thus it can be viewed as an invariant of two finite pseudometrics on the same ground set. For pseudometrics $d_1, d_2 : X \times X \to \mathbb{R}_{\ge 0}$ on $X = \{x_1, \dots, x_n\}$, define, for $i \in \{1, 2\}$ and $j, k \in \{1, \dots, n\}$,

$$D_i(j, k) := d_i(x_j, x_k) - \frac{1}{n}\sum_{l=1}^{n}\big(d_i(x_j, x_l) + d_i(x_k, x_l)\big) + \frac{1}{n^2}\sum_{l=1}^{n}\sum_{m=1}^{n} d_i(x_l, x_m),$$

where the summation indices l, m are taxon indices and are unrelated to the edge-length functions l_i. Then define

$$\mathrm{dCov}^2(d_1, d_2) := \frac{1}{n^2}\sum_{j=1}^{n}\sum_{k=1}^{n} D_1(j, k) D_2(j, k).$$

Here $D_i(j, k)$ is a centred distance-matrix entry and is unrelated to the directed-edge set D_i introduced in Section 2. We use the same symbol dCov^2 for the tree quantity and

the pseudometric quantity because their argument types distinguish them; Theorem 3 compares these two quantities explicitly.

Theorem 3 *For $X = \{x_1, \dots, x_n\}$ and any two edge-weighted X-trees $\mathcal{T}_1, \mathcal{T}_2$,*

$$\mathrm{dCov}^2(\mathcal{T}_1, \mathcal{T}_2) = \frac{3}{2}\,\mathrm{dCov}^2(d_{\mathcal{T}_1}, d_{\mathcal{T}_2}).$$

Proof A split $S = A \mid B$ defines the *split metric* d_S, where $d_S(x_1, x_2) = 0$ if x_1, x_2 lie on the same side and $d_S(x_1, x_2) = 1$ otherwise. For an edge-weighted tree,

$$d_{\mathcal{T}_i} = \sum_{uv \in E(T_i)} l_i(uv) d_{S_{uv}}.$$

Because metric distance covariance is bilinear in the two distance matrices, it suffices to consider two one-edge trees whose edges have length 1. Let their splits be $R_1 \mid R_2$ and $C_1 \mid C_2$, put $A_i^j := R_i \cap C_j$, and let r_i, c_j, a_i^j denote the corresponding cardinalities divided by n. Write $A = (a_i^j)_{i,j\in\{1,2\}}$ and $\Delta := \det A$. Then

$$\mathrm{dCov}^2(\mathcal{T}_1, \mathcal{T}_2) = Q(R_1 \mid R_2, C_1 \mid C_2) = 6\Delta^2,$$

so it remains to prove

$$\mathrm{dCov}^2(d_{R_1|R_2}, d_{C_1|C_2}) = 4\Delta^2.$$

For brevity, let $d_1 = d_{R_1|R_2}$ and $d_2 = d_{C_1|C_2}$. Given $i, j \in \{1, 2\}$, write $\bar{i}, \bar{j}$ for their complementary indices. If $x_k, x_l \in R_i$, then

$$D_1(k, l) = \frac{1}{n^2}\big(-2n|R_{\bar{i}}| + 2|R_1||R_2|\big) = -2r_{\bar{i}}^2,$$

whereas if x_k and x_l are separated by $R_1 \mid R_2$, then

$$D_1(k, l) = \frac{1}{n^2}\big(n^2 - n(|R_1| + |R_2|) + 2|R_1||R_2|\big) = 2r_1 r_2.$$

The corresponding formulas for D_2 are obtained by replacing R, r with C, c. Consequently,

$$\begin{aligned}\mathrm{dCov}^2(d_1, d_2) = 4\Bigg(&\sum_{i,j\in\{1,2\}} (a_i^j r_{\bar{i}} c_{\bar{j}})^2 - 2\sum_{i=1}^{2} a_i^1 a_i^2 r_{\bar{i}}^2 c_1 c_2 \\ &- 2\sum_{j=1}^{2} a_1^j a_2^j r_1 r_2 c_{\bar{j}}^2 + 2(a_1^1 a_2^2 + a_1^2 a_2^1) r_1 r_2 c_1 c_2\Bigg).\end{aligned}$$

Since the entries of A sum to 1,

$$a_i^j = r_i c_j + (-1)^{i+j}\Delta.$$

Substituting this relation for every entry, all terms except the multiples of Δ^2 cancel, and one obtains $\mathrm{dCov}^2(d_1, d_2) = 4\Delta^2$. □

The main result of [15] states that empirical distance covariance estimates the distance covariance of the underlying joint distribution when both vectors have finite first moments. This population quantity is nonnegative and vanishes exactly under independence. The result was extended from Euclidean spaces to metric spaces of strong negative type by Lyons [16]. For finite metrics this class agrees with strictly negative-type spaces, and tree metrics belong to it [17].

A pair $(\mathcal{T}_1, \mathcal{T}_2)$ of edge-weighted X-trees defines a discrete joint law by drawing U uniformly from X and setting $Z_i = \phi_i(U)$ for $i \in \{1, 2\}$. The mass of (v_1, v_2) is

$$\frac{|\phi_1^{-1}(v_1) \cap \phi_2^{-1}(v_2)|}{|X|}.$$

The random vertices are independent if and only if the trees are independent, by Lemma 4. Thus the preceding results justify using dCov^2 and dCor as measures of dependence between edge-weighted trees.

4.3 Strict positivity of Bergsma–Dassios τ^* under dependence

On path trees, the unweighted quartet covariance is directly related to the sign covariance introduced by Bergsma and Dassios [3] as a measure of dependence between ordinal random variables, extending Kendall's τ. They write t^* for a finite sample and τ^* for the expected value of t^* on four independent observations. Because this statistic is invariant under reversing either order, the weak order of a finite sample corresponds naturally to a labelled tree whose underlying graph is a path. Bergsma and Dassios proved that $\tau^* \geq 0$, with equality if and only if the variables are independent, when the joint law is discrete, jointly absolutely continuous, or a mixture of these two types, and conjectured the same conclusion for arbitrary bivariate laws. Drton, Han, and Shi [6] subsequently established both conclusions for random vectors with continuous margins, allowing joint laws that need not be absolutely continuous. We now establish both conclusions for arbitrary real-valued bivariate laws.

For the unnormalised Bergsma–Dassios convention and $z = (z_1, z_2, z_3, z_4)$, let r, s, t, u be pairwise distinct occurrence indices with $\{r, s, t, u\} = \{1, 2, 3, 4\}$; the notation $rs \mid tu$ divides the four occurrences into the two indicated pairs. Set

$$\begin{aligned} I_{rs|tu}(z) &:= \mathbf{1}\{\max(z_r, z_s) < \min(z_t, z_u) \text{ or } \max(z_t, z_u) < \min(z_r, z_s)\}, \\ a(z) &:= I_{13|24}(z) - I_{12|34}(z). \end{aligned}$$

The indicator $I_{rs|tu}$ and the kernel $a(z)$ are local to this subsection. They are unrelated to the earlier index sets $I_{t,u}$ and $I'_{t,u}$ and to the coefficient entries $a_{t,u}^{v,w}$. For four independent copies $(X^{(1)}, Y^{(1)}), \ldots, (X^{(4)}, Y^{(4)})$ of (X, Y), define

$$\tau^*(X, Y) := \mathbb{E}\Big[a(X^{(1)}, X^{(2)}, X^{(3)}, X^{(4)})a(Y^{(1)}, Y^{(2)}, Y^{(3)}, Y^{(4)})\Big].$$

Corollary 3 *Let (X, Y) be a real-valued random vector. If X and Y are not independent, then $\tau^*(X, Y) > 0$.*

Proof For the distribution-function discrepancy, write $D(x, y) := F(x, y) - F_X(x)F_Y(y)$, where $F(x, y) = P(X \leq x, Y \leq y)$, $F_X(x) = P(X \leq x)$, and $F_Y(y) = P(Y \leq y)$. This

local symbol D is unrelated to the directed-edge sets D_i and to the centred distance entries $D_i(j,k)$ used above. Introduce the unnormalised Blum–Kiefer–Rosenblatt discrepancy

$$\mathcal{B}(X,Y) := \int_{\mathbb{R}^2} D(x,y)^2 \, dF_X(x) \, dF_Y(y). \tag{8}$$

We prove the quantitative inequality

$$\tau^*(X,Y) \geq 2\mathcal{B}(X,Y). \tag{9}$$

Suppose first that X and Y have finite ordered marginal supports $x_1 < \cdots < x_m$ and $y_1 < \cdots < y_n$. We call each position (x_i, y_j) in the resulting marginal-support grid a cell, including positions having zero joint mass. Put $p_i = P(X = x_i) > 0$, $q_j = P(Y = y_j) > 0$, and, for $i < m$, $j < n$,

$$\Delta_{ij} := P(X \leq x_i, Y \leq y_j) - P(X \leq x_i)P(Y \leq y_j).$$

The symbols p_i and q_j are local marginal masses in this subsection and are unrelated to the coefficient families p and q from Section 2. Assume initially that all cell probabilities are rational. Replace each cell mass by the corresponding number of equally weighted labels, using one common finite label set for the two coordinates. Let T_X denote the labelled path tree whose ith vertex carries exactly the labels in row i, and let T_Y denote the labelled path tree whose jth vertex carries exactly the labels in column j. Thus they are labelled trees of the type defined in Section 2, with the replicated label set playing the role of the taxon set.

For a realised coordinate vector $z = (z_1, z_2, z_3, z_4)$, write $z_{(1)} \leq z_{(2)} \leq z_{(3)} \leq z_{(4)}$ for its order statistics. When $z_{(2)} < z_{(3)}$, define its *strict middle-cut partition*

$$\Pi_z := \big\{\{r : z_r \leq z_{(2)}\},\ \{r : z_r \geq z_{(3)}\}\big\};$$

both blocks have two occurrence indices. In the corresponding coordinate path tree, an edge separates the four occurrence indices into two pairs exactly when this strict inequality holds; in that case the induced positional partition is the unique partition Π_z. Repeated occurrences of one label have the same X- and Y-coordinates and therefore lie in the same block of every strict middle-cut partition. Equal X- and Y-partitions give one common partial quartet. If the partitions differ, their common refinement has four singleton blocks, so no label can be repeated and the displayed objects are two different full quartets. If either coordinate has no strict middle cut, the ordered quadruple contributes zero both to the quartet-covariance count and to the product kernel. This identifies the relevant events for all multiplicities and ties.

Let Π_X and Π_Y denote the strict middle-cut partitions of the four X- and Y-coordinates when they are defined, and set

$$C_4 := \{\Pi_X \text{ and } \Pi_Y \text{ are defined and } \Pi_X = \Pi_Y\},$$
$$D_4 := \{\Pi_X \text{ and } \Pi_Y \text{ are defined and } \Pi_X \neq \Pi_Y\}.$$

To determine the coefficients of $P(C_4)$ and $P(D_4)$ in the representation of τ^*, symmetrise over the four occurrence indices. Conditional on a common strict partition, a uniform permutation of the four occurrence indices sends it to $12 \mid 34$, $13 \mid 24$, or $14 \mid 23$ equally often. The corresponding values of a are $-1, +1, 0$, so the conditional mean of the product is $2/3$. Conditional on two distinct strict partitions, the six ordered pairs of distinct partitions occur equally often; two products are -1 and four are zero, so the conditional mean is $-1/3$. Exchangeability therefore gives

$$\tau^*(X,Y) = \frac{2}{3}P(C_4) - \frac{1}{3}P(D_4).$$

By the definition of quartet covariance,

$$Q(T_X, T_Y) = P(C_4) - \frac{1}{2}P(D_4) = \frac{3}{2}\tau^*(X, Y). \tag{10}$$

For $1 \le i < m$, let S_i^X be the split displayed by the edge of T_X between the vertices carrying x_i and x_{i+1}. For $1 \le j < n$, define S_j^Y analogously from T_Y. Here an endpoint mass means the normalized label mass $r_{t,u}(u)$ or $c^{v,w}(w)$ of the head vertex's label fibre in a directed-edge orientation. For the first edge, the orientation whose head is the vertex carrying x_i contributes p_i, and the reverse orientation contributes p_{i+1}. For the second edge the analogous two contributions are q_j and q_{j+1}. Thus the two orientation sums are $p_i + p_{i+1}$ and $q_j + q_{j+1}$, respectively. The corresponding quadrant determinant is

$$\det \begin{pmatrix} P(X \le x_i, Y \le y_j) & P(X \le x_i, Y > y_j) \\ P(X > x_i, Y \le y_j) & P(X > x_i, Y > y_j) \end{pmatrix} = \Delta_{ij}.$$

Thus $Q(S_i^X, S_j^Y) = 6\Delta_{ij}^2$, and Lemma 6 gives

$$Q(T_X, T_Y) \ge 3 \sum_{i=1}^{m-1} \sum_{j=1}^{n-1} (p_i + p_{i+1})(q_j + q_{j+1})\Delta_{ij}^2.$$

Together with (10),

$$\begin{aligned} \tau^*(X, Y) &\ge 2 \sum_{i=1}^{m-1} \sum_{j=1}^{n-1} (p_i + p_{i+1})(q_j + q_{j+1})\Delta_{ij}^2 \\ &\ge 2 \sum_{i=1}^{m-1} \sum_{j=1}^{n-1} p_i q_j \Delta_{ij}^2 = 2\mathcal{B}(X, Y). \end{aligned} \tag{11}$$

The atomic formula for $\mathcal{B}$ also has terms with $i = m$ or $j = n$, but $D(x_m, y_j) = D(x_i, y_n) = 0$, so those boundary terms vanish. Moreover, if the finite-support law is dependent, then some Δ_{ij} is nonzero. Since every displayed marginal mass is positive, the second inequality in (11) is then strict. Thus, for finite-support laws, equality $\tau^* = 2\mathcal{B}$ occurs only under independence, when both sides are zero.

The finite-support result extends to arbitrary, not necessarily rational, cell probabilities. Keep zero cells at zero and let $\mathcal{S} := \{(i, j) : P(X = x_i, Y = y_j) > 0\}$. For a cell $s = (i, j) \in \mathcal{S}$, abbreviate its probability by $p_s := P(X = x_i, Y = y_j)$. Choose $s_0 \in \mathcal{S}$ and, for every sufficiently large positive integer N, put

$$p_s^{(N)} = \frac{\lfloor N p_s \rfloor}{N} \quad (s \in \mathcal{S} \setminus \{s_0\}), \qquad p_{s_0}^{(N)} = 1 - \sum_{s \in \mathcal{S} \setminus \{s_0\}} p_s^{(N)}.$$

Every originally positive cell, and hence every occupied marginal row and column, remains positive for sufficiently large N. These rational tables converge to the original table. Both sides of (11) are continuous polynomial functions of the fixed table entries, so the inequality passes to the limit.

We now remove the finite-support assumption. For each integer $k \ge 1$, define the nested finite partition

$$\mathcal{P}_k := \{(-\infty, -k]\} \cup \{(j2^{-k}, (j+1)2^{-k}] : j \in \mathbb{Z},\ -k2^k \le j < k2^k\} \cup \{(k, \infty)\}.$$

Every endpoint of $\mathcal{P}_k$ is an endpoint of $\mathcal{P}_{k+1}$. Form the path tree for the X-coordinate from the cells of $\mathcal{P}_k$ having positive F_X-mass, in their inherited order, and define $g_k(x)$ to be the ordinal rank of the occupied cell containing x; set $g_k(x) = 0$ on the union of zero-F_X-mass cells. Define h_k analogously from the occupied F_Y-cells, and set $X_k = g_k(X)$, $Y_k = h_k(Y)$.

Almost surely, equal entries remain equal under every quantizer, while any distinct pair is eventually separated. The weak-order pattern of each sampled four-tuple—the collection of all pairwise relations $<$, $=$, and $>$ among its four coordinates—therefore eventually stabilises. The τ^* kernel is bounded, so dominated convergence gives

$$\tau^*(X_k, Y_k) \longrightarrow \tau^*(X, Y). \tag{12}$$

Let $r_k(x), s_k(y)$ denote the upper endpoints of the cells containing x, y, with a top tail interpreted as $+\infty$. These local endpoint maps are unrelated to the coefficient families r and s in Section 2. Extend D by

$$\bar{D}(+\infty, y) = \bar{D}(x, +\infty) = \bar{D}(+\infty, +\infty) = 0, \qquad \bar{D}(x, y) = D(x, y) \quad (x, y \in \mathbb{R}).$$

Because the cells are right closed, for $F_X \times F_Y$-almost every (x, y),

$$F_{X_k, Y_k}(g_k(x), h_k(y)) - F_{X_k}(g_k(x)) F_{Y_k}(h_k(y)) = \bar{D}(r_k(x), s_k(y)).$$

If X' and Y' are independent with laws F_X and F_Y, then

$$\mathcal{B}(X_k, Y_k) = \mathbb{E}\Big[\bar{D}(r_k(X'), s_k(Y'))^2\Big].$$

For finite x, y, eventually neither lies in a tail cell. Refinement makes the upper endpoints nonincreasing, and $0 \le r_k(x) - x < 2^{-k}$ and $0 \le s_k(y) - y < 2^{-k}$, so $r_k(x) \downarrow x$ and $s_k(y) \downarrow y$. The rectangles $(-\infty, r_k(x)] \times (-\infty, s_k(y)]$ decrease to $(-\infty, x] \times (-\infty, y]$. Continuity from above of the joint and marginal measures, followed by dominated convergence, implies

$$\mathcal{B}(X_k, Y_k) \longrightarrow \mathcal{B}(X, Y). \tag{13}$$

Applying (11) to (X_k, Y_k) and using (12) and (13) proves (9).

It remains to verify that $\mathcal{B}$ detects every failure of independence, including atomic and singular laws. For all $x, x', y, y' \in \mathbb{R}$,

$$|D(x, y) - D(x', y')| \le 2|F_X(x) - F_X(x')| + 2|F_Y(y) - F_Y(y')|. \tag{14}$$

Suppose $D(x_0, y_0) = d \neq 0$. Necessarily $F_X(x_0), F_Y(y_0) \in (0, 1)$. Choose $0 < \varepsilon < \min\{|d|/8, F_X(x_0), F_Y(y_0)\}$. The Borel sets

$$A := \{x \le x_0 : F_X(x_0) - \varepsilon < F_X(x) \le F_X(x_0)\},$$
$$C := \{y \le y_0 : F_Y(y_0) - \varepsilon < F_Y(y) \le F_Y(y_0)\}$$

have positive F_X- and F_Y-measure. Indeed, if $U = F_X(X)$ and $t = F_X(x_0) - \varepsilon$, then U need not be uniform when F_X has atoms, but it satisfies the super-uniform inequality $P(U \le t) \le t$. Hence $P(X \in A) \ge \varepsilon$, and similarly $P(Y \in C) \ge \varepsilon$. By (14), $|D(x, y)| > |d|/2$ on $A \times C$, so $\mathcal{B}(X, Y) \ge \frac{d^2}{4} P(X \in A) P(Y \in C) > 0$. Thus $\mathcal{B} = 0$ forces $D(x, y) = 0$ for all x, y. Equality of the joint and product measures on all lower-left rectangles implies equality on the Borel sigma-field. Conversely independence gives $D \equiv 0$. Therefore $\mathcal{B} = 0$ if and only if X and Y are independent. Dependence gives $\mathcal{B} > 0$, and (9) gives $\tau^*(X, Y) > 0$. If X and Y are independent, then the four-coordinate X-array and Y-array in the definition of τ^* are independent. Exchangeability gives zero expectation to each factor a, so $\tau^*(X, Y) = 0$. Thus Corollary 3 is equivalent to the zero characterisation stated in the abstract and introduction. □

5 Discussion

The central structural result is the nonnegative sum-of-squares representation of quartet covariance on finite labelled trees. Its full nonnegativity, combined with exact

quartet counting, yields the explicit finite-n quartet-distance bound and hence an independent derivation of the Bandelt–Dress asymptotic 2/3 upper bound. A coefficient estimate extracted from the first square family yields the path-tree inequality underlying the Bergsma–Dassios application. A further weighted identity identifies a quartet quantity with empirical distance covariance squared for induced tree pseudometrics. These are distinct consequences of the same finite-tree decomposition.

For bivariate laws, labelled path trees encode the weak-order patterns of finite distributions, and the coefficient estimate converts the local squares into a distribution-function discrepancy. Rational approximation and right-closed nested quantisation then pass from finite tables to arbitrary real bivariate laws. Because the four-sample kernels and the discrepancy are bounded, continuity from above and dominated convergence avoid moment, density, and continuity assumptions and treat atomic, mixed, and singular laws directly. Corollary 3 is a population identification result: no new sample-level limit theorem is proved here, and consequences for test consistency under atoms or singular components are not analysed.

Supplementary information

Supplementary Information 1: Lean 4 formalisation and reproducibility source code. The accompanying supplementary ZIP file contains the complete Lean 4 source closure for the machine-checked results cited in the manuscript, together with pinned Lean and mathlib versions, a manuscript-to-Lean correspondence table, an axiom audit, integrity hashes, and licence information.

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Use of artificial intelligence

The authors used OpenAI's GPT-5.6 Sol with the ultra reasoning-effort setting to assist in completing the proofs of Lemma 6 and Corollary 3, and in constructing the accompanying formalisation in Lean 4 and mathlib. The authors checked and revised all mathematical arguments and formal code and take full responsibility for the contents of the manuscript and the formalisation.

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